🧠 INCREASED INTRACRANIAL PRESSURE (ICP)

Increased ICP occurs when the pressure inside the skull rises, compressing brain tissue and impairing blood flow. It's a **neurological emergency** and must be recognized early to prevent **brain herniation** and death.

📄 KEY TERMINOLOGY

- **Intracranial** – (intra = within; cranial = skull) Inside the skull
- **Pressure** – Force exerted by fluids or tissues in a confined space
- **Herniation** – (herni = to protrude) Brain tissue is forced from one compartment to another — often fatal
- **Cerebral perfusion pressure (CPP)** – Blood flow to brain tissue
- **Cushing's triad** – Classic signs of high ICP affecting the brainstem

💡 NORMAL RANGES

- **Normal ICP**: 5–15 mmHg
- **↑ ICP**: >20 mmHg (sustained)
- **CPP = MAP – ICP** → Goal CPP ≥ 60 mmHg

🚨 CAUSES OF INCREASED ICP

- **Head trauma** (bleeds, swelling)
- **Brain tumor**
- **Stroke** (especially hemorrhagic)
- **Hydrocephalus** (fluid buildup in ventricles)
- **Meningitis, encephalitis** (infections)
- **Hypoxia, hypercapnia** (↑ CO_2 → cerebral vasodilation)

⚠️ EARLY SIGNS OF ↑ ICP

- **Headache** (worse with coughing, straining)

- **Vomiting** (often without nausea; “projectile”)
- **Change in LOC** (confusion, restlessness)
- **Blurred or double vision (diplopia)**
- **Pupil changes** (sluggish, unequal)

🚨 LATE SIGNS OF ↑ ICP

Cushing’s Triad:

- **↑ Systolic BP** with **widened pulse pressure**
- **Bradycardia** (slow heart rate)
- **Irregular respirations** (Cheyne-Stokes or apnea)

Posturing:

- **Decorticate** (arms flexed to chest; cerebral cortex injury)
- **Decerebrate** (arms extended; brainstem injury—worse)

Seizures

Fixed, dilated pupils

Coma

🧪 DIAGNOSTIC TESTS

- **CT or MRI** – Visualize swelling, mass effect
- **Avoid lumbar puncture** if ↑ ICP is suspected (can cause herniation!)

🛏️ NURSING CARE: ICP MANAGEMENT (NCLEX MUST-KNOW)

📍 POSITIONING:

- **Head of bed (HOB) at 30°**
- Keep head **midline and neutral** (no flexion or rotation)

💨 OXYGENATION & CO_2:

- Maintain **$PaCO_2$ 35–38 mmHg** (↑ CO_2 causes vasodilation → ↑ ICP)
- Avoid hypoxia and hypercapnia

⚕️ GENERAL INTERVENTIONS:

- Maintain **quiet, low-stimulation environment**
- Limit suctioning and avoid straining
- Prevent fever (↑ metabolism → ↑ ICP)
- **Strict I&O** – monitor for SIADH or diabetes insipidus
- **Seizure precautions**

💊 MEDICATIONS FOR ICP

Osmotic diuretics:

- **Mannitol** – pulls fluid from brain tissue into blood
- Monitor **serum osmolality and kidney function**
- **Hypertonic saline (3%)** – reduces brain swelling
- **Corticosteroids** (e.g., dexamethasone) – only for tumors, not trauma
- **Sedatives & analgesics** (reduce agitation and metabolic demand)
- **Anticonvulsants** (e.g., phenytoin) – prevent or treat seizures
- **Antipyretics** (e.g., acetaminophen) – control fever

🔍 NCLEX TEST STRATEGY TIPS

- **Change in LOC = #1 early sign** of ↑ ICP
- **Cushing's triad = Late, ominous sign** of impending herniation
- **Never** position flat or trendelenburg → worsens ICP
- **Avoid lumbar puncture** if ICP is suspected!
- Monitor for **SIADH** or **DI** post-neuro trauma

💡 MNEMONIC: PRESSURE

P – Position HOB 30°, neutral head
R – Respiratory (keep CO_2 down, suction cautiously)
E – Elevated temp → control fever

S – Systems monitoring (neuro checks, ICP, CPP)
S – Straining avoided (no coughing, sneezing, Valsalva)
U – Unconscious patient → protect airway
R – Rx: Mannitol, hypertonic saline
E – Edema managed